

When We See Five Clusters, Are There Really Five Biological Populations?

What clustering algorithms actually tell us in single-cell analysis

A computational perspective on representation, neighborhood graphs, community detection, resolution, and biological interpretation

The question behind the colorful map

When we see five clusters in a single-cell dataset, are there really five biological populations? Or did our computational pipeline produce five groups because we asked an algorithm to partition the data?

Single-cell RNA sequencing has given us extraordinary maps of cellular diversity.

A typical paper may begin with tens of thousands of cells and end with a beautifully colored UMAP containing clusters labeled:

0, 1, 2, 3, 4.

We identify marker genes.

We assign names.

Cluster 0 becomes “CD4 T cells.”

Cluster 1 becomes “CD8 T cells.”

Cluster 2 becomes “B cells.”

Soon, the computational partition begins to look like a direct description of the tissue itself.

But something important happened between the original expression matrix and those final biological labels.

We made choices.

Many of them.

We chose how to normalize the data, which genes to retain, how many principal components to use, how to define similarity between cells, how many neighbors each cell should have, how those neighbor relationships should be weighted, which community-detection objective should be

optimized, and at what resolution the graph should be partitioned.

Only after all of that did we obtain the clusters.

The central idea

A clustering algorithm does not operate directly on biology.

It operates on a computational representation of biology.

The clusters we obtain are therefore influenced by both the underlying biological structure and the mathematical choices used to represent that structure.

This does not make clustering invalid.

It means we should be precise about what clustering has actually shown.

1. Two Questions That Sound Similar but Are Not

Suppose we have a dataset containing thousands of cells.

There are two fundamentally different questions we could ask.

Question 1: Does discrete structure actually exist?

Does the dataset contain naturally separated populations with meaningful boundaries between them?

Question 2: How should the data be partitioned?

Given a mathematical definition of similarity and a clustering objective, what is a good way to divide the observations into groups?

These questions are related, but they are not equivalent.

Many clustering algorithms primarily solve the second problem.

They optimize a partition.

They do not first perform a universal statistical test asking whether the biological system genuinely consists of discrete populations.

The distinction becomes obvious with one of the simplest clustering algorithms.

2. The k -Means Thought Experiment

Suppose we run k -means with

$$K = 5.$$

The algorithm searches for five centroids,

$$\mu_1, \mu_2, \mu_3, \mu_4, \mu_5,$$

and assigns each observation to one of them while minimizing within-cluster variation.

The classical objective is

$$\operatorname{argmin}_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2.$$

The optimization problem is therefore:

What is the best five-way partition of these observations?

It is not:

Do five naturally occurring populations exist?

If we give k -means $K = 5$, it will return five groups.

Even if the original data form one continuous cloud.

A simple computational experiment

Imagine 1,000 observations drawn from one smooth distribution with no obvious gaps.
Now run

$$k\text{-means}, \quad K = 5.$$

The algorithm will still return

$$C_1, \quad C_2, \quad C_3, \quad C_4, \quad C_5.$$

The five labels demonstrate that the observations can be divided into five regions according to the k -means objective.

They do not, by themselves, demonstrate that five natural populations existed before clustering.

Single-cell analysis usually uses more sophisticated graph-based clustering methods such as Leiden or Louvain.

But the conceptual lesson survives.

A mathematical partition and a biological population are different objects.

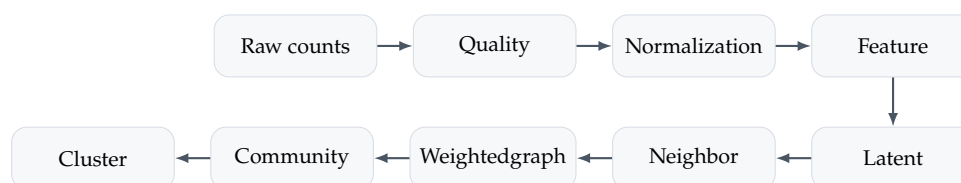
3. What Actually Happens Before Single-Cell Clustering?

A simplified single-cell workflow is often written as:

Counts $\rightarrow$ PCA $\rightarrow$ Clustering.

That summary is convenient, but it hides most of the computational modeling.

A more realistic conceptual pipeline looks like this.



The crucial point is that Leiden usually does not receive the raw gene-expression matrix.

It receives a graph.

And even that graph is not simply “the k -nearest-neighbor graph” in the most naive sense.

There is usually another modeling step in between.

4. Nearest Neighbors Are Only the Beginning

Suppose each cell has been represented in a d -dimensional latent space, often using principal components.

For cell i ,

$$z_i = (PC_{i1}, PC_{i2}, \dots, PC_{id}).$$

We calculate distances between cells and identify a set of nearest neighbors:

$$N_k(i).$$

At this stage, we have answered a local question:

Which cells are closest to cell i under the representation and distance metric we selected?

But many modern single-cell pipelines do not simply give Leiden a binary graph saying:

$$A_{ij} = \begin{cases} 1, & j \in N_k(i), \\ 0, & \text{otherwise.} \end{cases}$$

Instead, the nearest-neighbor relationships are transformed into a **weighted similarity graph**.

That distinction matters.

4.1. Seurat: Shared Nearest Neighbors

In a typical Seurat workflow, nearest neighbors are first identified and can then be transformed into a shared nearest-neighbor graph.

The basic intuition is simple.

Suppose cells i and j are near one another.

If their neighborhoods also contain many of the same cells, their relationship receives stronger support.

A simple measure of neighborhood overlap is the Jaccard index:

$$J(i, j) = \frac{|N_k(i) \cap N_k(j)|}{|N_k(i) \cup N_k(j)|}.$$

Thus two cells can be connected not merely because one appears in the other's nearest-neighbor list, but because their local neighborhoods overlap substantially.

The resulting SNN graph becomes the object on which graph clustering is performed.

4.2. Scanpy: Weighted Neighborhood Connectivities

Scanpy follows a related but mathematically different approach.

Its standard neighbor construction produces both distances and weighted neighborhood connectivities.

Under the commonly used UMAP-based neighborhood construction, local distances are converted into fuzzy connectivity strengths.

Conceptually, a pair of cells may therefore have an edge weight

$$w_{ij} \in [0, 1],$$

where larger values indicate stronger local connectivity under the chosen neighborhood model.

Leiden can then use these weighted connectivities as its adjacency structure.

The important point is not that Seurat and Scanpy use exactly the same weighting rule.

They do not.

The important point is that in both cases, another modeling layer exists between

nearest-neighbor search

and

community detection.

A step that is easy to overlook

Before Leiden starts, we have already decided:

- how cells are represented,
- how distance is measured,
- how many neighbors define “local,”
- how nearest-neighbor relationships are transformed into weighted edges,
- whether weak edges are retained or pruned,
- and which graph is eventually passed to the clustering algorithm.

Leiden therefore clusters a **constructed similarity graph**.
It does not cluster biological reality directly.

This strengthens the main argument considerably.

The graph itself is already a model.

5. The Similarity Graph Is a Scientific Assumption

Consider the computational chain

expression $\rightarrow$ latent coordinates $\rightarrow$ distance $\rightarrow$ neighbors $\rightarrow$ edge weights.

Every arrow contains assumptions.

For example, changing the number of principal components changes the coordinates on which distances are computed.

Changing the metric from Euclidean distance to cosine distance changes what “nearby” means.

Changing k alters the scale at which local neighborhoods are defined.

Changing graph weighting alters the strength of relationships passed to Leiden.

Changing integration or batch-correction methods can change the latent geometry before the graph is even constructed.

Therefore,

the graph is not merely found

the graph is constructed

from the data under a particular set of modeling decisions.

That does not mean the graph is arbitrary.

A good graph can capture extremely meaningful biological structure.

But it is important to remember what kind of object it is.

6. What Does Leiden Actually Do?

Once the weighted graph has been created, we still have another computational question:

How should this graph be divided into communities?

Leiden is an optimization algorithm for finding high-quality communities in a graph.

Importantly, there is not one universal mathematical objective called “the Leiden objective.”

Leiden can optimize different graph partition quality functions.

One commonly used formulation is a resolution-adjusted modularity objective.

A representative form is

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \gamma \frac{k_i k_j}{2m} \right] \delta(c_i, c_j),$$

where

$$A_{ij}$$

represents the edge weight between cells i and j ,

$$k_i$$

represents the weighted degree of cell i ,

$$m$$

summarizes total graph connectivity,

$$c_i$$

is the community assignment of cell i ,

and

$$\delta(c_i, c_j) = \begin{cases} 1, & c_i = c_j, \\ 0, & c_i \neq c_j. \end{cases}$$

The parameter

$$\gamma$$

controls the scale at which community structure is favored.

The algorithm seeks a partition that scores well under the selected objective.

Notice what does not appear anywhere in this equation:

T cell,

macrophage,

tumor state,

or

biological population.

The algorithm sees a graph.

The biological interpretation comes afterward.

7. Leiden Is an Algorithm, Not a Single Objective Function

This distinction is worth emphasizing.

In some implementations, Leiden is used with modularity-related objectives such as the Reichardt-Bornholdt configuration model.

Another important option is the Constant Potts Model, or CPM.

A simplified CPM objective can be expressed as

$$Q_{\text{CPM}} = \sum_{i,j} (A_{ij} - \gamma) \delta(c_i, c_j),$$

with the precise formulation depending on conventions used for the graph and edge counting.

The mathematical interpretation differs from standard modularity-based formulations.

This matters because different quality functions encode different ideas of what constitutes a good community.

They can also behave differently with respect to graph size and the classical resolution-limit problem of modularity.

An important computational distinction

“Leiden” describes the community-detection algorithm.

It does not uniquely specify the objective function being optimized.

The graph representation, edge weights, quality function, resolution parameter, and optimization procedure are separate modeling choices.

In practice, software defaults also matter.

A pipeline that says only

Run Leiden

may hide several assumptions that affect the final partition.

8. The Resolution Parameter Changes the Biological Granularity

Suppose the same cell graph is clustered repeatedly.

At one resolution we obtain

4 clusters.

At another we obtain

8 clusters.

At a finer resolution we obtain

15 clusters.

Which answer is correct?

The tempting response is to search for the one “true” number of clusters.

But biological systems often have structure at multiple scales.

Consider peripheral blood.

At a coarse level, we might describe:

Lymphoid cells and Myeloid cells.

At another level:

T cells, B cells, NK cells, Monocytes.

At finer resolution:

CD4 T, CD8 T, Naive B, Memory B, Classical monocytes, Non-classical monocytes.

At still finer resolution, transcriptional state may begin to dominate:

Activated, Cycling, Interferon-responsive, Stress-associated.

The tissue did not change when the resolution parameter changed.

Our computational description changed.

A better question

Instead of asking only

“What is the correct number of clusters?”

it is often more informative to ask

“At what biological scale do we want to describe this system?”

Higher resolution is not automatically better.

Lower resolution is not automatically better.

The appropriate scale depends on the scientific question and on whether the resulting subdivisions are stable and biologically interpretable.

9. Biology Really Can Be Hierarchical

This is not merely a computational inconvenience.

Biology itself is often hierarchical.

For example,

Immune cells

can be divided into

Lymphoid and Myeloid.

Lymphoid cells can be divided further:

T cells, B cells, NK cells.

T cells can be divided into:

CD4 T and CD8 T.

CD4 T cells can be subdivided again into states such as:

Naive, Memory, Activated.

Several resolutions may therefore all describe legitimate aspects of the same biological system.

The mistake is assuming that one particular resolution necessarily reveals a unique and absolute partition of nature.

10. But Biology Is Also Often Continuous

The problem becomes even more interesting when the underlying biology is not naturally discrete.

Many biological processes are continuous.

Differentiation is an obvious example:

Stem cell → Progenitor → Intermediate state → Mature cell.

Transcriptional programs can change gradually along such a trajectory.

There may be no exact point at which one state suddenly becomes another.

Imagine cells distributed smoothly along a developmental axis.



Now partition this smooth continuum into five communities:

$$C_1 \mid C_2 \mid C_3 \mid C_4 \mid C_5.$$

Those labels may be useful summaries.

They may help compare gene expression or isolate regions of the trajectory.

But the boundaries between them do not automatically imply five naturally discrete cell types.

A major interpretational trap

A boundary produced by a clustering algorithm is not automatically a biological boundary. Sometimes the boundary exists because the optimization procedure must assign cells to communities even when the underlying variation is gradual.

This is one reason methods such as trajectory inference, pseudotime analysis, diffusion-based representations, RNA velocity, and continuous latent-variable models are useful.

They allow us to ask whether the dominant biological structure may be better represented as a continuum.

11. A Cluster Is Not Automatically a Cell Type

Suppose Leiden returns cluster 7.

What exactly is cluster 7?

It could represent a genuine cell type.

But it could also reflect:

- a cell subtype,
- a transient cell state,
- an activation program,
- differentiation,
- cell-cycle phase,
- interferon response,
- metabolic state,
- cellular stress,
- treatment response,
- donor-specific variation,
- sequencing batch,
- library complexity,

- mitochondrial signal,
- doublets,
- ambient RNA contamination,
- or an algorithmic subdivision of continuous variation.

Therefore,

Computational cluster $\neq$ Biological cell type

unless additional evidence supports that interpretation.

Cluster labels are mathematical outputs.

Cell-type labels are biological hypotheses.

12. The UMAP Can Make the Clusters Feel More Real Than They Are

After clustering, we usually color a UMAP by cluster identity.

Suddenly, the groups become visual objects.

One cluster may occupy an island.

Another may form a narrow branch.

A third may appear separated from everything else.

The visual impression is powerful.

It can also encourage circular reasoning.

Consider a simplified workflow:

Latent representation $\rightarrow$ Neighborhood graph.

The graph is then used for community detection.

The same or closely related neighborhood structure is also used to construct the UMAP representation.

Conceptually,

Neighborhood structure $\rightarrow \begin{cases} \text{Community detection} \\ \text{UMAP visualization} \end{cases}$

The cluster labels and their UMAP positions are therefore not two independent experiments.

They are related outputs built from closely related representations of local cell similarity.

A UMAP island is not independent validation

Seeing cluster 5 occupy a visually separated UMAP region is useful for visualization. It is not, by itself, independent evidence that cluster 5 represents a discrete biological population.

This does not mean UMAP is misleading.

It means that UMAP should be interpreted for what it is: a low-dimensional representation designed largely around preservation of local neighborhood relationships.

13. Marker Genes Do Not Completely Solve the Problem

A natural response is:

“But the cluster has marker genes.”

Surely that means the population is real?

Not necessarily.

Imagine a gene whose expression changes continuously along a biological trajectory:

$$g(x) = \alpha + \beta x.$$

Now divide the continuum into two computational groups:

Early and Late.

A differential-expression test may identify g as strongly different between those groups.

That result can be completely statistically valid.

The late cells really do express more g .

But this does not prove that a sharp biological boundary exists between ‘early’ and ‘late.’

The boundary was introduced by the partition.

A subtle but important distinction

Differential expression can demonstrate that two computationally defined groups differ. It does not necessarily demonstrate that the boundary separating those groups represents a naturally discrete biological transition.

This is particularly important when cluster markers are calculated using exactly the same cells that were first partitioned based on their expression profiles.

The marker analysis helps characterize the partition.

It does not independently prove that the partition was inevitable.

14. What About Silhouette Scores?

Cluster quality can also be evaluated geometrically.

One common measure is the silhouette coefficient.

For observation i , let

$$a(i)$$

be its average distance to observations within its own cluster.

Let

$$b(i)$$

be the smallest average distance from i to observations in another cluster.

Then

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

Values near 1 indicate that an observation is substantially closer to its own cluster than to neighboring clusters.

Values near 0 suggest overlapping structure.

Negative values can indicate that an observation may be closer to another cluster.

This is useful information.

But what does a high silhouette score actually establish?

It tells us that the clusters are geometrically well separated under the chosen representation and distance metric.

It does not tell us that each cluster is necessarily a distinct biological cell type.

Geometric separation $\neq$ Biological identity

The distinction between computational validity and biological validity appears again.

15. Does This Mean Clustering Invents Biology?

No.

That conclusion would go too far.

Real biological populations can produce extremely strong and reproducible community structure.

B cells and monocytes, for example, may occupy very different transcriptional neighborhoods because they are genuinely different cell populations.

Clustering can recover such structure remarkably well.

The point is not that clusters are artificial.

The point is that the existence of a cluster label alone is insufficient to establish what kind of biological entity the cluster represents.

A useful conceptual model is:

Observed cluster = Biological structure+Measurement+Representation+Graph construction+Clustering choice

When the biological signal is strong, many reasonable pipelines may recover nearly the same population.

When the structure is weak, continuous, hierarchical, or confounded by technical variation, the computational choices become much more influential.

16. One Dataset, Two Analysts, Two Clusterings

Imagine two analysts receive exactly the same expression matrix.

Analyst A uses:

3,000 highly variable genes,

30 principal components,

and

$k = 20$.

Analyst B uses:

2,000 highly variable genes,

50 principal components,

and

$$k = 40.$$

Perhaps they also choose different integration methods or clustering resolutions.

The resulting graphs may differ.

The resulting partitions may differ.

Neither analyst necessarily made an error.

They constructed different computational representations of the same measurements.

The scientific question is then:

Which biological conclusions survive reasonable analytical choices?

Parameter sensitivity is information

If a proposed cell population disappears after a tiny change in the number of PCs, neighbors, graph weighting, or resolution, that instability is not merely an annoyance. It is evidence about how strongly the data support that population.

17. How Do We Decide Whether a Cluster Is Biologically Meaningful?

There is no single universal test.

Confidence comes from multiple pieces of evidence pointing toward the same interpretation.

17.1. 1. Stability across reasonable parameter choices

Does the population remain recognizable when we slightly vary:

number of HVGs,

number of PCs,

k ,

resolution,

or other reasonable analysis settings?

A biologically strong structure should ideally not depend on one extremely specific parameter combination.

This does not require every cell to retain exactly the same label.

Real boundaries can be fuzzy.

The question is whether the underlying population remains recognizable.

17.2. 2. Reproducibility across biological replicates

Does the cluster occur across donors, samples, or experiments?

A cluster found almost entirely in one donor may reflect interesting biology.

But it could also reflect:

batch, sample quality, library preparation, processing differences.

Replication provides an important layer of evidence.

17.3. 3. Coherent marker programs

Biological populations are often supported by coordinated transcriptional programs.

A convincing T-cell population should ideally be supported by multiple biologically related genes.

A cluster defined almost entirely by:

mitochondrial genes,

ribosomal genes,

or

generic stress genes

requires a different interpretation.

The question is not merely which genes have the smallest adjusted p -values.

The question is whether the marker set describes a coherent cellular identity or state.

17.4. 4. Relationship to technical covariates

Cluster identity should always be compared with variables such as:

donor,

batch,

library size,

mitochondrial fraction,

cell-cycle score,

doublet score.

If one technical variable nearly predicts cluster identity, the cluster may be describing the experimental pipeline rather than the intended biology.

17.5. 5. Stability under subsampling or perturbation

What happens if a fraction of cells is removed?

What happens if the graph is reconstructed?

What happens if clustering is repeated with different random seeds or reasonable parameter perturbations?

Stable structure should tend to reappear.

Unstable subdivisions deserve more cautious biological labels.

17.6. 6. Agreement with the biological question

Not every transcriptionally distinguishable group deserves a new cell-type name.

Sometimes the scientifically relevant interpretation is:

cell state,

rather than

cell type.

Sometimes the correct conclusion is:

continuum,

rather than

discrete population.

The biological ontology should follow the evidence, not the cluster numbering.

17.7. 7. Independent validation

The strongest evidence often comes from information that the clustering algorithm never used.

Examples include:

- protein measurements,
- flow cytometry,
- spatial localization,
- histology,
- lineage information,
- functional perturbation,
- known surface markers,
- independent cohorts,
- orthogonal molecular assays.

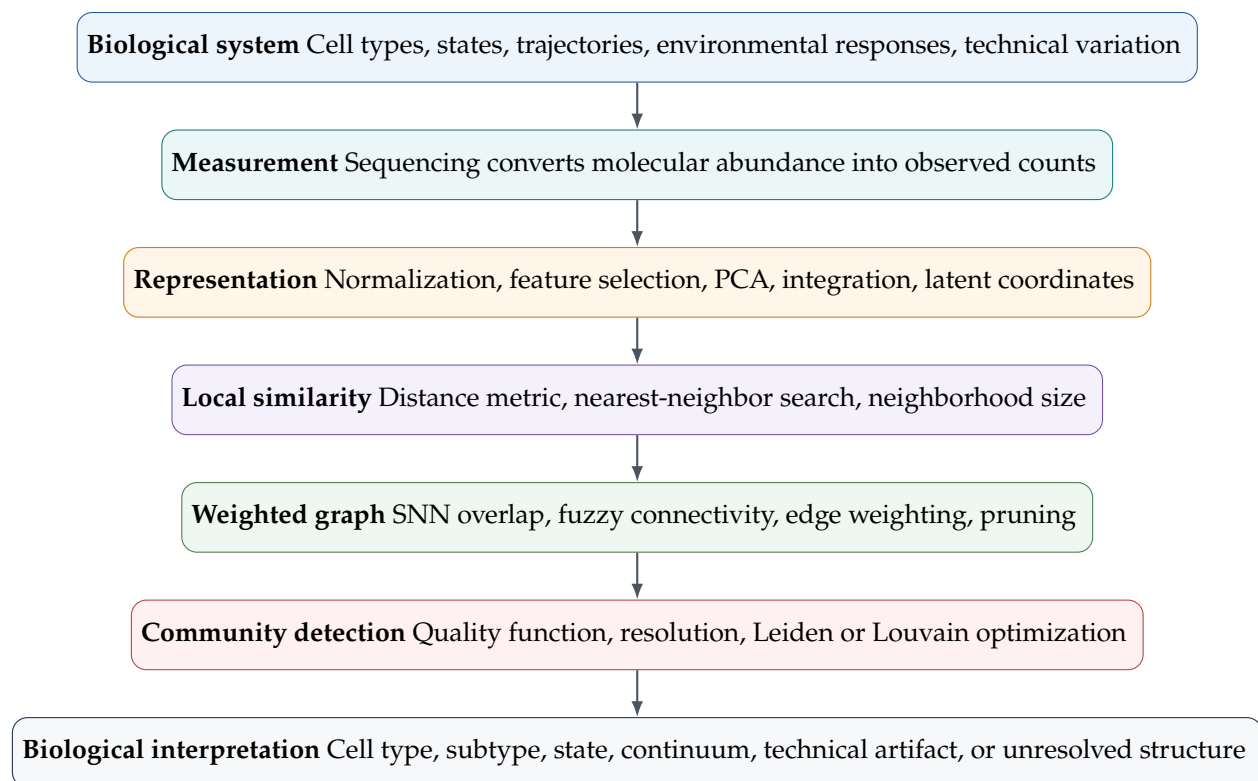
Why independent evidence matters

Evidence becomes especially powerful when it does not arise from the same expression matrix that was used to construct the clusters.

Independent information can test whether the computational partition corresponds to something reproducible outside the clustering procedure itself.

18. A Better Mental Model of Single-Cell Clustering

The full process can be thought of as a sequence of transformations.



Every arrow introduces assumptions.

That does not invalidate the result.

It identifies where interpretation enters the analysis.

19. Perhaps the Most Important Equation Is Not an Algorithm

We often focus on individual methods:

PCA, UMAP, Leiden.

But the final clustering is better understood as a function:

$$C = f(X, T, F, Z, D, k, W, Q, \gamma),$$

where

X = measured expression data,

T = transformations and normalization,

F = selected features,

Z = latent representation,

D = distance metric,

k = neighborhood size,

W = graph weighting scheme,

Q = community quality function,

and

γ = resolution parameter.

The final cluster assignment is therefore not simply

$$C = f(\text{biology}).$$

It is the result of biology passing through a computational pipeline.

The computational lesson

Algorithms do not simply reveal structure.
They reveal structure through a mathematical representation and an optimization objective.
Understanding the representation and the objective is part of understanding the biological result.

20. Questions Worth Asking Before Naming a New Cell Population

When a new cluster appears, a useful checklist is:

Before calling cluster 7 a new cell type

- What features drove the separation?
- Which latent dimensions contributed to the neighborhood structure?

- Is the cluster stable across reasonable values of k and resolution?
- Does it remain recognizable under reasonable changes in feature selection or the number of PCs?
- Is it represented across independent biological samples?
- Is cluster membership strongly associated with batch or technical QC variables?
- Does the population have a coherent marker program?
- Are those markers describing identity, activation, stress, or cell cycle?
- Could the apparent groups instead represent a continuous trajectory?
- Does independent evidence support the proposed biological interpretation?

These questions do not weaken clustering.

They make the biological inference stronger.

21. So What Can We Actually Claim When Leiden Returns Five Clusters?

A cautious computational statement would be:

Under the selected feature representation, neighborhood definition, weighted graph construction, community-detection objective, and resolution, the cells can be partitioned into five communities.

That is a legitimate result.

But this stronger statement requires additional evidence:

The tissue contains exactly five discrete biological populations.

These two claims should not be treated as equivalent.

The first is a computational result.

The second is a biological interpretation.

22. The Deeper Lesson Goes Beyond Single-Cell RNA-seq

This problem is not unique to single-cell biology.

The same principle appears whenever algorithms transform complex data into categories.

In patient stratification, clustering can divide a disease continuum into molecular subtypes.

In microbiome studies, samples can be grouped into community states.

In cancer genomics, tumors may be divided into expression subtypes.

In proteomics, samples may form apparent molecular classes.

In each case, the algorithm provides a useful partition.

But the scientific question remains:

Does the partition correspond to a naturally discrete system, or is it a useful summary of continuous variation?

That question cannot be answered by cluster labels alone.

23. Final Thought

Single-cell analysis has given us extraordinarily detailed maps of tissues.

But every colorful clustering figure contains at least two layers.

The first is computational:

How did we represent and partition the data?

The second is biological:

What does that partition mean?

Confusing these layers can turn an analytical choice into a biological conclusion.

Keeping them separate makes the science stronger.

The central takeaway

When an algorithm returns five clusters, it has not automatically demonstrated that five biological populations exist.

It has demonstrated that, under a particular representation, definition of local similarity, graph-construction method, community objective, and parameter setting, the observations can be organized into five computational communities.

Determining whether those communities correspond to five distinct biological populations is a separate scientific problem.

So perhaps the first question after clustering should not be:

What are these five cell types?

Perhaps it should be:

Why did the algorithm see five groups?

And does the biology agree?

References and Further Reading

1. Traag, V. A., Waltman, L., and van Eck, N. J. *From Louvain to Leiden: guaranteeing well-connected communities*. Scientific Reports, 2019.
2. Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. *Fast unfolding of communities in large networks*. Journal of Statistical Mechanics: Theory and Experiment, 2008.
3. Waltman, L. and van Eck, N. J. *A smart local moving algorithm for large-scale modularity-based community detection*. European Physical Journal B, 2013.
4. McInnes, L., Healy, J., and Melville, J. *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. 2018.
5. Wolf, F. A., Angerer, P., and Theis, F. J. *SCANPY: large-scale single-cell gene expression data analysis*. Genome Biology, 2018.
6. Hao, Y. et al. *Integrated analysis of multimodal single-cell data*. Cell, 2021.
7. Kiselev, V. Y., Andrews, T. S., and Hemberg, M. *Challenges in unsupervised clustering of single-cell RNA-seq data*. Nature Reviews Genetics, 2019.
8. Luecken, M. D. and Theis, F. J. *Current best practices in single-cell RNA-seq analysis: a tutorial*. Molecular Systems Biology, 2019.
9. Seurat reference documentation: *Shared nearest-neighbor graph construction and Cluster determination*. Accessed 2026.
10. Scanpy reference documentation: *Neighbor graph construction and Leiden community detection*. Accessed 2026.